Niacin (Vitamin B_3) Metabolism and Pellagra

NAD and NADPH are hydrolyzed in the intestinal lumen to form **nicotinamide**. Nicotinamide can be converted to **nicotinic acid** by intestinal bacteria or absorbed directly into the bloodstream. Both nicotinamide and nicotinic acid are transported to the liver, kidneys, and intestines, where they are reconverted to NAD and NADPH. These coenzymes serve as hydrogen donors and acceptors in oxidation-reduction reactions critical for the metabolism and synthesis of carbohydrates, fatty acids, and proteins.

Epidemiology

Deficiency of **niacin (vitamin B_3)** leads to **pellagra**. In 1735, Gaspar Casal described a group of poor peasants in northern Spain suffering from a skin disorder he termed "mal de la rosa," characterized by a reddish, glossy rash on the dorsum of the hands and feet. These individuals subsisted primarily on maize and rarely consumed fresh meat.

Pellagra became endemic in the southern United States in the early 1900s due to a diet largely composed of cornbread, molasses, and pork fat. The disease remains prevalent in parts of the world such as South Africa, China, and India, where corn and maize are dietary staples. However, the niacin in corn is present in a bound form that is not bioavailable unless released through alkaline hydrolysis (e.g., nixtamalization with lime water).

Jowar, a type of millet consumed in parts of India, contains niacin but also high levels of **leucine**, which inhibits the conversion of **tryptophan** to niacin. In contrast, pellagra is uncommon among Mexican populations despite their maize-based diet because traditional preparation involves soaking maize in lime water, which liberates bound niacin.

Gastrointestinal disorders can predispose individuals to pellagra due to impaired absorption of tryptophan and niacin. Conditions associated with malabsorption include: - Jejunoileitis - Gastroenterostomy - Prolonged diarrhea - Anorexia nervosa - Chronic colitis - Ulcerative colitis - Cirrhosis - Crohn disease - Subtotal gastrectomy

Hartnup disease, a rare autosomal recessive disorder, results in defective intestinal and renal transport of neutral amino acids, including tryptophan. This leads to malabsorption and subsequent pellagra-like symptoms in childhood.

Alcoholism increases the risk of pellagra due to both poor dietary intake and malabsorption. Similarly, overly restrictive diets—seen in eating disorders, food faddism, or perceived food allergies—can lead to niacin deficiency.

Patients with **carcinoid syndrome** are at increased risk because excessive tryptophan is shunted toward serotonin production (up to 60% instead of the normal 1%), reducing substrate availability for niacin synthesis.

Medications Associated with Pellagra

Certain drugs can induce pellagra by interfering with niacin metabolism: - **Isoniazid**: Competes with NAD due to structural similarity and impairs pyridoxine (vitamin B_6) function, which is essential for tryptophan-to-niacin conversion. - **5-Fluorouracil**: Inhibits the conversion of tryptophan to niacin. - **6-Mercaptopurine**: Inhibits NAD phosphorylase, reducing NAD production. - Other implicated agents: Phenytoin, chloramphenicol, azathioprine, sulfonamides, and some antidepressants.

Clinical Findings

Pellagra is classically characterized by the **"four Ds"**: 1. **Dermatitis** 2. **Diarrhea** 3. **Dementia** 4. **Death**

The dermatitis is **photosensitive** and typically begins as erythematous, painful, or pruritic patches. Over several days, the skin becomes edematous and may develop vesicles and bullae. These lesions can rupture, leading to crusted erosions, or evolve into brown, scaly patches.